

PANDA Component Description: *rs232_buffered*

Overview

The *rs232_buffered* component implements a RS-232 port with configurable input- and output-buffer. The baudrate is defined through a generic. The component has the ability to generate interrupts if a databyte is sent or received, the send buffer is empty, the receive buffer is full or if a specified character was received.

Generics

Name	Type	Default	Description
clk_frequency	integer	10'000'000	Clock frequency in Hz
baudrate	integer	9'600	Baudrate in baud
recbuf_size	integer	128	Receive buffer size (only powers of 2 allowed)
sndbuf_size	integer	128	Send buffer size (only powers of 2 allowed)

External Ports

Name	Type	Width	Description
rx	std_logic	1	Serial input data
tx	std_logic	1	Serial output data

Registers

Data Register

Name: DAT
Address: Base + 0x00

Name	DAT
R/W	R/W
Reset Value	0x00

Operations on the *DAT* register access the write/receive buffers. Read operations on the *DAT* register return the next byte from the receive buffer. Write operations on the *DAT* register write one byte to the send buffer.

A write operation on *DAT* clears the *SND* bit in *INT*, a read operation on *DAT* clears the *REC* and the *CHAR* bit in *INT*.

If *DAT* is written while the send buffer is full, the write operation does not have any effect. The written data will not be sent.

Received Character Register

Name: RCHR

Address: Base + 0x01

Name	RCHR
R/W	R/W
Reset Value	0x00

The *RCHR* register holds a character value. If this character is received, the *CHAR* bit in *INT* is set. If *CHAR* in *CFG* is also set, an interrupt is generated. This can be used to detect the reception of a full command if *RCHR* is set to the end-sign of a the command (usually a carriage return).

Configuration Register

Name: CFG

Address: Base + 0x02

Name	-	SFUL	REMP	CHAR	SEMP	RFUL	REC	SND
R/W	-	R	R	R/W	R	R	R/W	R/W
Reset Value	0	0	1	0	0	0	0	0

SND:	'1'	→	Enable interrupt if data is sent (<i>SND</i> in <i>INT</i> is set)
	'0'	→	Disable interrupt if data is sent
REC:	'1'	→	Enable interrupt on receiving data (<i>REC</i> in <i>INT</i> is set)
	'0'	→	Disable interrupt on receiving data
RFUL:	'1'	→	Enable interrupt if receive buffer is full (<i>RFUL</i> in <i>INT</i> is set)
	'0'	→	Disable interrupt if receive buffer is full
SEMP:	'1'	→	Enable interrupt if send buffer is empty (<i>SEMP</i> in <i>INT</i> is set)
	'0'	→	Disable interrupt if send buffer is empty
CHAR:	'1'	→	Enable interrupt if <i>RCHR</i> is received (<i>CHAR</i> in <i>INT</i> is set)
	'0'	→	Disable interrupt if <i>RCHR</i> is received
REMP:	'1'	→	Receive buffer is empty
	'0'	→	Receive buffer is not empty
SFUL:	'1'	→	Send buffer is full
	'0'	→	Send buffer is not full

Interrupt Register

Name: INT

Address: Base + 0x03

Name	-	-	-	CHAR	SEMP	RFUL	REC	SND
R/W	-	-	-	R/W	R	R	R/W	R/W
Reset Value	0	0	0	0	1	0	0	0

SND: '1' → Data send over RS-232 completed since last clear
 '0' → No data send over RS-232 completed since last clear

REC: '1' → Received data over RS-232 since last clear
 '0' → Received no data over RS-232 since last clear

RFUL: '1' → Receive buffer is full
 '0' → Receive buffer is not full

SEMP: '1' → Send buffer is empty
 '0' → Send buffer is not empty

CHAR: '1' → Received *RCHR* over RS-232 since last clear
 '0' → Received no *RCHR* over RS-232 since last clear

CHAR, *REC* and *SND* can be cleared by writing a '0' to them. Writing a '1' to these bits does not affect their state. They can only be cleared but not set by software. *SEMP* and *RFUL* are read only bits, write operations to these bits do not have any effect.

REC and *CHAR* are also cleared if *DAT* is read, the *SND* bit is cleared if *DAT* is written.